SEMICONDUCTOR / PLASMA / PROCESS MONITORING

Plasma diagnostics and monitoring evaluation.

Optical emission spectroscopy, model-conditional tomography and controlled monitoring stress tests, separating synthetic evidence from measured-data calibration.

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50% to 13.56%
Mean false-positive rate
Transferred recalibrated

61.09% to 99.39%
Mean unconditional recall
Same future evaluation frames

37,814
Distinct future frames
3 sessions / 6 reused roles

Public measured N2 study; 512 dataset-labelled normal reference frames. See pp. 6-7.

Research-to-engineering focus. I applied an established laboratory CT-OES framework, evaluated its assumptions, and developed a portfolio extension to test synthetic state changes and alarm robustness. A separate public measured N2 study shows why a near-perfect AUC need not yield reliable transferred thresholds.

SKILLS DEMONSTRATED / MEASUREMENT

OES · Gaussian fitting · inverse problems · radiation-trapping assumptions · Python analysis

SKILLS DEMONSTRATED / EVALUATION

Parent-grouped synthetic comparisons · PCA/SVM · correlated-noise tests · threshold transfer · abstention cost

Not a production-fab qualification or real-fault detector. No actual yield uplift, fab process control, or Ar-to-N_2 transfer has been demonstrated. Data provenance is kept explicit on each page.

01 / OPTICAL EMISSION DIAGNOSTICS

From measured spectra to spatial maps

The project began with measured optical-emission data.

312 viewing channels Measured Ar spectra	**2,808 line fits** 9 emission lines
CT reconstruction 31 × 31 grid	**Plasma interpretation** Model-conditional

118
quality-review flags
Not predictive failures

31 × 31
spatial reconstruction grid

50°
code-defined LoS span
−25° to +25°

My CT-OES contribution at SNU

I applied the laboratory's existing Python CT-OES framework, organised the channel inputs and exclusions, examined Gaussian fitting and quality flags, and interpreted regularised reconstructions. I compared relative spatial trends with supplied Langmuir-probe measurements while tracking the differences in resolution, perturbation and calibration assumptions.

SIGNAL PROCESSING

Constrained multi-line Gaussian fitting · overlap handling · fit review · background and calibration assumptions.

INVERSE RECONSTRUCTION

Ray geometry · nonnegative regularisation · projection-versus-field error · edge artefacts.

Why the quality-control count matters

Of 2,808 spectral fitting tasks, 118 received quality-review flags. That is an **analysis workload and QC metric**, not a model accuracy score or a claim that 118 devices would fail.

> **Attribution boundary.** The CT framework predated my placement; applying it, investigating reconstruction results and creating the follow-on portfolio evaluations are my contributions. I do not claim to have invented the laboratory reconstruction platform.

[S1] SNU research summary and PlasmaSense run_summary.json (v0.10.0); geometry is a code configuration, not independent metrology of the original apparatus. Original chamber raw data are not reproduced here.

02 / RECONSTRUCTION VALIDATION

Reconstruction accuracy and uncertainty

An inverse solution must be evaluated in the reconstructed field, with model assumptions and rejected regions visible.

Reconstruction accuracy cannot be inferred from a small projection residual

29.23%
field relative L2 error
Noiseless same-forward-model phantom

2.176 eV
model-conditional T_e median
412 accepted pixels

0.45 m
assumed effective path
Not an independently measured path

The noiseless phantom check showed that even when a solver reproduced line-of-sight projections, the recovered **spatial field could still be substantially wrong**. Limited-angle geometry and regularisation make field fidelity a separate validation question.

Temperature and pixel acceptance

Pixel status	Count	What it means
Accepted model-conditional T_e pixels	412	Used for the reported 2.176 eV median
Metastable fits rejected or at bounds	303	Not presented as reliable temperature estimates
Outside or invalid measurement support	245	Not assigned reconstructed physical conditions
Temperature root not bracketed	1	No accepted root under chosen assumptions
Total grid	**961**	31 × 31 pixels

> **Do not conflate historical and new results.** An earlier Donnelly/Dasgupta-based OES analysis reported a 1.42 eV representative value; the later recovered-BSR dataset, path and spectral-response assumptions yield 2.176 eV in PlasmaSense. They are **different model-conditioned estimates**, not two verified observations of the same absolute physical quantity.

[S1] run_summary.json: atomic_dataset, geometry, diagnostics and phantom_relative_l2_error; results/tables/plasma_diagnostics.csv. No independently validated absolute electron density is claimed.

03 / PHYSICS-INFORMED EVALUATION

Synthetic state-change evaluation

The simulator tests a deliberately limited scientific question and keeps synthetic labels separate from observed operating outcomes.

What PlasmaSense actually simulates

The extension applies **reduced-physics interventions** to an experimental Ar emission template. Changes to effective electron energy and relative electron density produce labelled synthetic responses under declared excitation and observation assumptions.

Measured anchor
One Ar condition

Imposed states
Energy · density · local variation

Synthetic signals
Shared geometry and template

Detector comparison
Held-out parent groups

Four detector approaches

Comparator	Feature or training strategy	Purpose
Original PCA	Original projection/signal representation	Baseline unsupervised anomaly response
Mean-feature PCA	Mean signal features for each line	Test reduced feature sensitivity
Noise-aware mean PCA	Normal-noise augmentation during development	Test robustness to independent channel noise
Physics-trained SVM	Supervised examples from imposed model states	Test controlled-state discrimination

SPLIT DISCIPLINE

Synthetic children of a common parent remain in one partition; learning and normal-reference scores are fixed before changed-noise evaluation.

CRITICAL LIMITS

The generator is not driven by real RF power, pressure or gas recipe. Uniform relative density scaling is confounded with optical gain.

What the study answers. It compares recognition of stated synthetic changes and stress sensitivity. It does **not** demonstrate real fab fault detection, optical hardware robustness or universally superior physics-trained learning.

[S2] Saved grouped evaluations. Demo inventory: 2,520 replay cases / 10,080 detector decisions; these are interface verification counts, not independent experiments.

04 / COVARIANCE STRESS

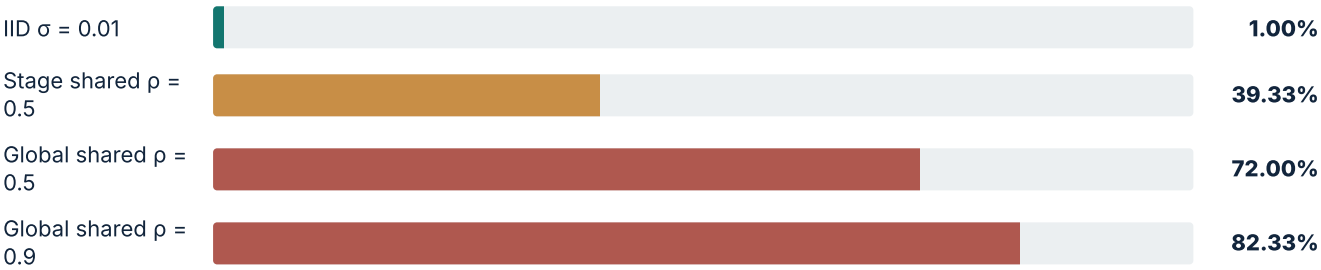
Sensitivity to correlated noise

A lower false-alarm rate under independent noise was not a general guarantee of reliable monitoring.

Noise robustness: where a promising method failed

Under the stated Maxwellian-response synthetic regime, normal-noise augmentation reduced IID false alarms in one stress setting. But correlated noise, plausible when measurement channels share optical or acquisition disturbances, exposed a major failure mode.

MEAN FPR · NOISE-AWARE MEAN-FEATURE PCA



For a separate IID high-noise condition, original PCA FPR was 100% and noise-aware mean PCA was 8% (mild-change recall 62.22%). The improved IID result did not transfer to these correlated-noise shifts. [S2]

Condition	Observed comparison	Important trade-off
IID σ 0.01	Original PCA FPR 2.67%; noise-aware 1.00%	Mild-change recall: 62.56% → 57.78%
IID σ 0.03	Original PCA FPR 100%; noise-aware 8.00%	Noise-aware mild recall 62.22%
Stage-shared ρ 0.5; σ 0.01	Noise-aware FPR 39.33%	Calibrated IID robustness breaks under covariance change
Global-shared ρ 0.5 / 0.9	Noise-aware FPR 72% / 82.33%	Shared-channel effects overwhelm the normal reference

Decision. Do not call a model “noise robust” merely because IID tests improve. Retest the correlation structure of normal measurement errors and report both false alarms and the sensitivity lost on subtle state changes.

[S2] state_improvement_families and robustness comparison outputs; means over five simulator seeds around one experimental anchor. Not confidence intervals, independent physical experiments or measured industrial fault rates.

05 / MEASURED-DATA ALARM TRANSFER

Threshold transfer across sessions

A public measured dataset tests calibration transfer on unchanged future evaluation populations.

Separate measured evidence. To test threshold transfer, I used a public N_2 controlled-injection OES dataset containing **52,393 spectra from three physical sessions**. This is not the Ar state generator and does not validate Ar-to-N_2 fault transfer.

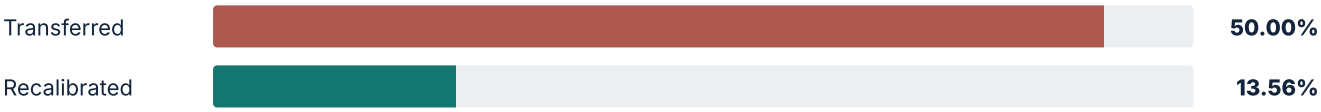
0.9989
global mean AUC for author-index LR
Six reused role permutations

37,814
distinct matched future frames
Calibration evaluation

512
dataset-labelled normal frames
Reference acquired before evaluated frames

Same future frames, different threshold policy

MEAN FALSE POSITIVE RATE



MEAN LEAK RECALL



Interpretation. A nearly perfect score ranking did not define a transferable decision threshold. A target-session labelled-normal reference improved alarm operating points, but one session still produced 33.22% false alarms.

Critical denominator: three physical sessions were reused in six train/test role permutations; averages are descriptive. The broader AUC is 0.998915, while the matched calibration-subset AUC is 0.998360. Do not quote both as one exact evaluation. Acquisition frames and early leaks before the 512-frame reference were excluded. [S3]

06 / DECISION POLICY

False alarms and missed events

The best-looking false-positive metric is not automatically the safest or most operationally useful policy.

A cleaner alarm chart can hide a dangerous loss

Policy	Mean FPR	Mean unconditional recall	Leak frames withheld
Recalibrated, no guard	13.56%	99.39%	0%
Add 3% support guard	9.97%	52.56%	46.83%

MEAN UNCONDITIONAL LEAK RECALL



AT FIRST GLANCE

False alarms fall by 3.59 percentage points: 13.56% → 9.97%.

THE HIDDEN COST

Nearly half of leak frames are withheld; unconditional recall falls by about 46.83 percentage points.

Engineering decision: no automatic veto. I retained the guard as a **review/uncertainty signal**, rather than using it to suppress a primary alarm. A high accepted-only recall does not account for real events deferred by the system.

Limitations that remain

Valve-operation and temporal effects were not causally isolated from leak labels. Channel 127 cannot be assigned an unambiguous wavelength due to axis/column alignment differences. The 512-frame normal reference is **dataset-labelled**, not externally proven by independent fab sensors. Threshold choices were explored during development on inspected data; no newly held-out factory validation is claimed.

[S3] external_calibration/summary.csv; external_guard/summary.csv and metrics.csv. All FPR/recall/guard figures above are means across reused role permutations. Recall is unconditional and counts withheld leak frames as not alarmed.

07 / OWNERSHIP AND RELEVANCE

My analysis and engineering decisions

Clear ownership and negative results are part of the technical evidence, not caveats hidden at the end.

What I contributed

Phase	My documented work
Diagnostic foundation	Applied the SNU CT-OES framework; reviewed channel ordering, Gaussian fitting and spectral QC.
Inverse-model checks	Kept geometry explicit, investigated regularised field errors and temperature assumption sensitivity.
Physics-informed evaluation	Defined reduced-physics interventions, parent-grouped splits and four detector comparisons.
Noise robustness	Evaluated fixed detector/calibration settings under IID, stage-shared and global-shared channel noise.
Measured benchmark	Tested threshold transfer and normal-reference calibration on public N_2 sessions and matched future frames.
Decision and communication	Measured guard abstention cost; documented reasons to keep review flags rather than auto-suppress alarms.

PROJECT OWNERSHIP

PlasmaSense is an independent portfolio extension from supervised research. I developed the analysis framing, evaluated the results and documented the engineering decisions.

WHAT IS NOT CLAIMED

A new CT framework at SNU, independently operated ALD/XPS tools, real fab fault data, validated yield improvement or a production digital twin.

How this relates to semiconductor process engineering

These results motivate a practical workflow: inspect spectrometer/optics QC for signal failures; obtain a labelled or operationally confirmed reference when transfer shifts; and investigate persistent alerts with tool logs and wafer metrology **before** changing a recipe. These are recommended checks, not deployed fab procedures.

PLASMA DIAGNOSTICS · PYTHON · SPECTRAL FITTING · CT-OES · PCA / SVM · NOISE ROBUSTNESS · MODEL VALIDATION

08 / ROLE ALIGNMENT & SOURCES

Relevant skills and references

A concise explanation of scope, limitations and decision-making is as important as the underlying numbers.

Skills demonstrated

PROCESS DIAGNOSTICS

My work connects plasma spectroscopy with monitoring reliability. I applied CT-OES analysis to 312 measurement channels, then tested how model assumptions, channel noise and threshold transfer affect the trustworthiness of a monitoring decision. I would bring that diagnostic discipline to process and yield investigations.

MODEL EVALUATION

My background combines low-temperature plasma diagnostics, numerical reconstruction and careful experimental interpretation. In my PlasmaSense extension I tested controlled synthetic changes and identified correlated-noise failure conditions; this motivates independent verification before drawing conclusions about manufacturing process health.

ADVANCED MANUFACTURING / EQUIPMENT

I enjoy translating difficult measurement and modelling results into practical engineering checks. My projects cover quantitative spectral fitting, limited-angle reconstruction and measured-data alarm calibration, with explicit attention to false positives, missed events and model boundaries.

Methods and source roles
Schulze et al. (2008), J. Phys. D 41, 065206. Common-upper ratios and escape factors. DOI 10.1088/0022-3727/41/6/065206.
Chouteau et al. (2024), PSST 33, 075016. Eq. (8) metastable balance; Eqs. (10)-(11) relative ne. DOI 10.1088/1361-6595/ad5d11.
Donnelly (2004), J. Phys. D 37, R217. Historical Fig. 5 thick direct-plus-cascade input. DOI 10.1088/0022-3727/37/19/R01.
Dasgupta et al. (published 1999), Phys. Rev. A 61, 012703. Fig. 6 theoretical 1s5-to-2p channels. DOI 10.1103/PhysRevA.61.012703.
Boffard et al. (2012), J. Phys. D 45, 045201. Osaka 420.1/419.8 nm method only. DOI 10.1088/0022-3727/45/4/045201.
Public N2 data: Youngjun Yuk, Mendeley Data v1, DOI 10.17632/9gy9v3h4mp.1, CC BY 4.0. Separate retrospective label-assisted evaluation.
Donnelly primary lineage: Schappe et al. (1994), PRA 50, 444; Chutjian & Cartwright (1981), PRA 23, 2178. These historical method references are distinct from the later demo's model inputs and do not certify its numerical results.

Contact
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Version 1.2 / 31 October 2026. Graduate research portfolio. Charts reproduce reported results; no new experimental or operational performance certification.

Methods and evaluation scope

Measured diagnostics vs synthetic replay

The diagnostic workload contains 9 lines x 312 channels. The saved detector replay uses 8 line features: 696.54, 706.72, 738.40, 750.38, 763.51, 800.62, 801.48 and 842.47 nm. These are different analysis scopes; the replay does not display all diagnostic fits.

Frozen / mean-feature PCA

Frozen means fitted models and normal calibration scores remain fixed during noise stress. Mean-feature PCA uses eight line-wise means of log(signal/reference) across 312 channels.

Noise correlation and response assumptions

Rho denotes imposed same-line channel log-noise correlation: within a 78-channel stage or across all 312 channels. Maxwell refers to a Maxwellian electron-energy response. EEDF shape variation is a sensitivity study; its energy parameter is not automatically Te.

Same model / different operating policy

The measured benchmark uses an author-index single-channel logistic classifier. Target-session normal-reference calibration changes its threshold; the support guard withholds decisions. The displayed N2 policy comparison does not retrain a model.

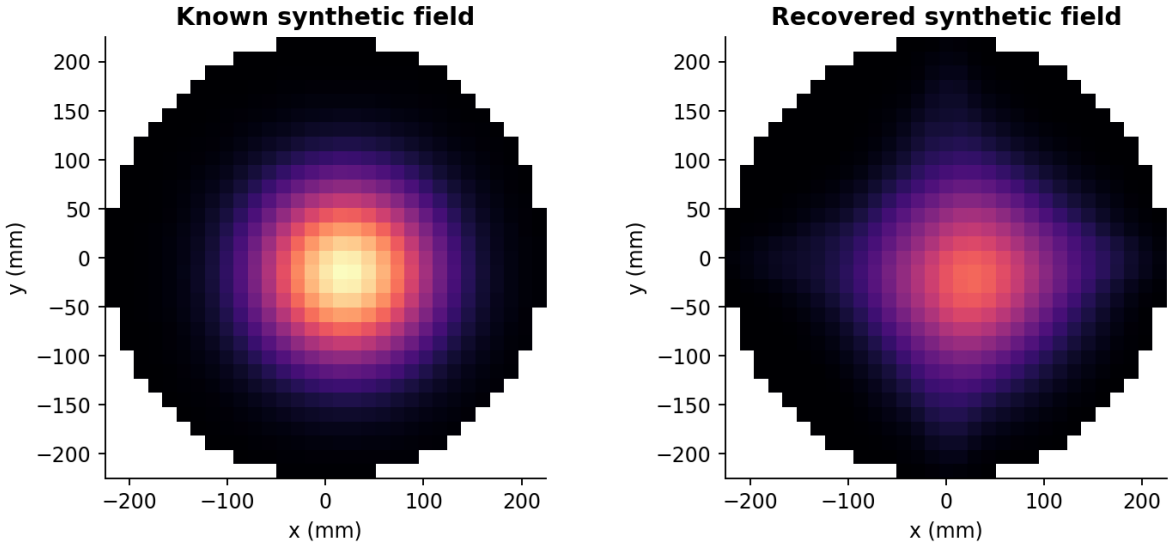
Reproducibility and evaluation scope

The reported replay comprises 2,520 saved cases and 10,080 detector decisions. These are consistency-check counts, not independent experiments or held-out scientific validation. This application edition presents results in the portfolio; no interactive lab or downloadable research data is included.

Supporting results

Selected results / technical review

Phantom validation · same forward model · synthetic, not an experiment



29.23% field relative L2 error in a noiseless, same-forward-model phantom. This synthetic check reveals smoothing and field distortion; a small projection residual does not establish spatial accuracy.

NOISE STRUCTURE CHANGES THE OUTCOME

At σ = 0.01, noise-aware mean PCA FPR rose from 1% under independent noise to 39.33% with stage-shared correlation ρ = 0.5 and 72% with global-shared ρ = 0.5. Five reused seeds, assumed covariance and fixed models/thresholds. Physics-trained SVM was not uniformly superior.

SCOPE OF THIS APPLICATION EDITION

This portfolio presents previously reported aggregate results and selected figures. The compact website includes the Highlights and technical PDF; research CSVs, source code and the interactive lab are not included.

Boundaries: N_2 valve/time effects remain unresolved. Ar synthetic and N_2 measured studies are distinct. Absolute n_e and pressure/power/gas-ratio prediction are unsupported. Laboratory CT engine: existing framework; SNU analysis and portfolio evaluation: my work.

Sources: retained PlasmaSense evaluation results; Youngjun Yuk, Mendeley Data v1, DOI 10.17632/9gy9v3h4mp.1 (CC BY 4.0). Public dataset attribution does not imply that private laboratory data are distributed.